

MECT: MIXTURE OF EXPERTS WITH CNN-TRANSFORMER NETWORK FOR SPEAKER VERIFICATION

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ABSTRACT

In this paper, we propose MECT, a speaker verification model that integrates the Mixture-of-Experts (MoE) mechanism into a CNN-Transformer backbone with optimized block structure and stacking scheme. Specifically, we investigated four MoE variants that span utterance-level and frame-level granularity with dense and sparse routing strategies. The MoE mechanism proves to be effective over the baseline without MoE with only a small increase in parameters. We further scale MECT to a series of model sizes, all maintaining compact parameters and low computational complexity. In particular, MECT-B2 achieves state-of-the-art performance on VoxCeleb1 and delivers strong results on CN-Celeb, demonstrating its effectiveness across diverse datasets. In addition, we establish a streaming inference paradigm through causal retraining, which maintains strong performance at a chunk size of 100ms.

Index Terms— speaker verification, MoE, streaming inference

1. INTRODUCTION

Speaker verification (SV) aims to determine whether two speech segments are from the same speaker. In recent years, deep neural network architectures for SV have evolved rapidly. Representative models include 1D convolution TDNN-based frameworks such as ECAPA-TDNN [1, 2] and Next-TDNN [3], CNN-based architectures such as ResNet [4, 5] and its variants [6, 7] including DF-ResNet and Gemini DF-ResNet [8, 9], and CNN-TDNN hybrid structures such as CAM++ [10] and ECAPA2 [11]. Transformer-based networks have also been investigated [12–15]. ReDimNet [15] introduces a reshape-dimension strategy to effectively integrate CNN, TDNN, and Transformer modules, while ReDimNet2 [16] further improves performance through time-pooled dimension reshaping. Self-supervised pre-trained models (PTMs) have also been applied to speaker verification, with w2v-BERT 2.0-based SV [17] achieving particularly strong performance through fine-tuning.

Mixture-of-Experts (MoE) [18] has been widely adopted in large-scale Transformer-based systems, improving scalability through dynamic expert routing. In speaker verification, MoE has only been applied to fine-tuning PTMs [19, 20], where it adaptively combines representations from different layers of the pretrained model. However, MoE has not yet been explored in fully supervised SV.

We propose MECT, a fully supervised speaker verification model that introduces the MoE mechanism. Four MoE structures are designed with different expert granularities and routing strategies: utterance-level Dense MoE, utterance-level Sparse MoE,

frame-level Dense MoE, and frame-level Sparse MoE, each replacing the first linear projection in the Transformer feed-forward block. MECT adopts a CNN-Transformer architecture based on the Reshape Dimension strategy of ReDimNet, with a redesigned stacking scheme and network structure for better performance. The main contributions of this paper are summarized as follows:

1) We propose MECT, the first MoE-based fully supervised speaker verification model, achieving state-of-the-art performance on Vox1-O, Vox1-E, and Vox1-H with minDCF of 0.012, 0.026, and 0.048, while maintaining low model complexity. 2) We investigate four MoE structures and demonstrate performance gains over the baseline across datasets, with frame-level dense MoE excelling on VoxCeleb and sparse Top-K MoE on CN-Celeb. 3) We establish a streaming inference paradigm for SV through causal retraining, achieving competitive performance at a chunk size of 100ms.

2. METHODS

This section describes the architecture of MECT. We first introduce the basic building block, then detail the MoE layer as the core component and a key contribution of this work. Next, we present the stacking scheme for constructing MECT models at different scales to accommodate diverse deployment requirements. Finally, we describe the streaming speaker embedding extraction scheme.

2.1. Basic Block of MECT

As illustrated in Figure 1, a MECT block consists of a CNN and a Transformer equipped with MoE. The CNN and Transformer are connected via the reshape and time-pooled dimension-reshaping methods adopted from ReDimNet and ReDimNet2 [15, 16]. Beyond MoE, our block design also differs from ReDimNet in several aspects that contribute to further performance gains.

Residual connection. We adopt ResNet blocks for the CNN, but place the shortcut addition after BatchNorm and ReLU rather than before, as shown in the CNNBlock module of Figure 1. This design consistently yields better performance in our experiments.

Position encoding. We employ a two-layer 1D convolution as the position encoding, as illustrated in the PosEncConv of Figure 1.

Downsampling and channel expansion. Both temporal downsampling and channel expansion are handled by convolutions within the ResNet block, rather than by a separate convolution layer before it as in ReDimNet. This design simplifies the module structure and leads to more stable training.

Block connection. We connect blocks via a simple shortcut rather than the dense connections used in ReDimNet. In our experiments, dense connections did not improve performance while increasing computational overhead.

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<https://github.com/ant-research/AntSpeaker>

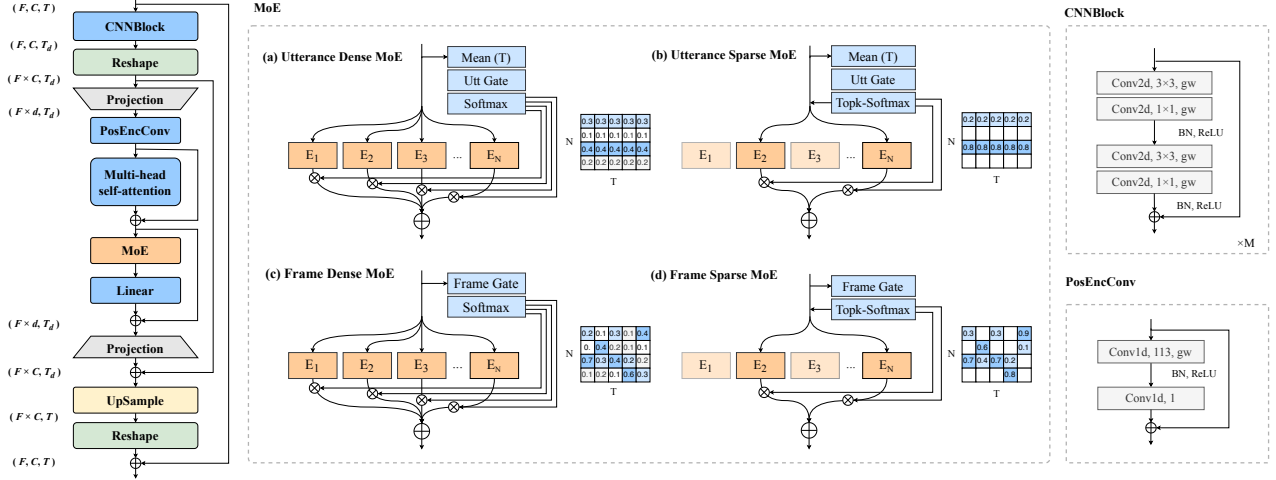


Fig. 1: The architecture of MECT Block. Left shows the overall block, middle the four MoE variants, and right the CNN and PosEncConv modules.

2.2. Mixture-of-Experts layer

We explore four MoE variants that differ in aggregation granularity (utterance-level vs. frame-level) and routing strategy (Dense vs. Sparse), as shown in Figure 1.

Let $\mathbf{x} \in \mathbb{R}^{T \times d}$ denote the input feature sequence to the MoE layer, where T and d are the temporal length and feature dimension, respectively. Let $\mathbf{x}_t \in \mathbb{R}^d$ denotes a single frame.

2.2.1. Frame-level MoE

In the Dense MoE, the gating network is a linear layer parameterized by $\mathbf{W}_g \in \mathbb{R}^{d \times N}$, where N denotes the number of experts. Its output $\mathbf{G}_t$ serves as the routing weights for each expert at frame t . Each expert is a linear layer parameterized by $\mathbf{W}_e^i$, and $f(\cdot)$ denotes the activation function. The output of the MoE layer, $\mathbf{y}_t$, is computed as

$$\mathbf{G}_t = \text{Softmax}(\mathbf{x}_t \mathbf{W}_g), \quad \mathbf{y}_t = \sum_{i=1}^N \mathbf{G}_t^i \cdot f(\mathbf{x}_t \mathbf{W}_e^i). \quad (1)$$

In the Sparse MoE, only the Top- K experts are activated for routing. $\mathcal{S}_t$ and $\mathbf{h}_t$ denote the indices and the corresponding logits of the selected experts, respectively. It is formulated as follows:

$$\mathcal{S}_t, \mathbf{h}_t = \text{TopK}(\mathbf{x}_t \mathbf{W}_g, K), \quad (2)$$

$$\mathbf{G}_t = \text{Softmax}(\mathbf{h}_t), \quad \mathbf{y}_t = \sum_{i \in \mathcal{S}_t} \mathbf{G}_t^i \cdot f(\mathbf{x}_t \mathbf{W}_e^i). \quad (3)$$

2.2.2. Utterance-level MoE

In the utterance-level variants, the gating decision is made based on the mean-pooled utterance representation $\bar{\mathbf{x}} = \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t$, producing a single set of routing weights shared across all frames. The utterance-level Dense MoE is formulated as follows:

$$\mathbf{G} = \text{Softmax}(\bar{\mathbf{x}} \mathbf{W}_g), \quad \mathbf{y}_t = \sum_{i=1}^N \mathbf{G}^i \cdot f(\mathbf{x}_t \mathbf{W}_e^i). \quad (4)$$

And the utterance-level Sparse MoE is formulated as follow:

$$\mathcal{S}, \mathbf{g} = \text{TopK}(\bar{\mathbf{x}} \mathbf{W}_g, K), \quad (5)$$

$$\mathbf{G} = \text{Softmax}(\mathbf{g}), \quad \mathbf{y}_t = \sum_{i \in \mathcal{S}} \mathbf{G}^i \cdot f(\mathbf{x}_t \mathbf{W}_e^i). \quad (6)$$

Table 1: Architecture of MECT.

Layer name	Structure	Output
Input	-	(1, 80, T)
Head	Conv2D, stride 1	(C, 80, T)
Stage1	Block $\times$ 2, stride 1	(C, 80, T)
Stage2	Block $\times$ 4, stride (2,1),(1,2) $\times$ 3	(2 $\times$ C, 40, T)
Stage3	Block $\times$ 5, stride (2,1),(1,2) $\times$ 4	(4 $\times$ C, 20, T)
Stage4	Block $\times$ 3, stride (2,1),(1,2) $\times$ 2	(8 $\times$ C, 10, T)
Fusion	Weight Dense	(80 $\times$ C, T)
Pooling	ASTP	80 $\times$ C
Projection	Linear	192
Loss	SphereFace2	Num. Speakers

2.3. MECT architecture

Table 1 details the MECT architecture. Each stage stacks multiple MECT blocks, where the first block applies frequency downsampling, and the remaining blocks apply temporal downsampling, both via convolutional stride. Outputs of all stages are fused via a weighted dense layer before pooling.

We provide four MECT model sizes controlled by two parameters: the input channel size C and the number of ResBlocks per block M , with a fixed Transformer hidden dimension $d = 64$. A1 and B1 use $M = 2$, while A2 and B2 use $M = 3$; A1 and A2 use $C = 16$, while B1 and B2 use $C = 32$.

We use 80-dimensional mean-normalized log Mel filter-bank features extracted with a 25 ms window and 10 ms frame shift over 20–7600 Hz at a sampling rate of 16 kHz. Attentive Statistics Pooling (ASTP) [21] is employed for utterance-level aggregation, and SphereFace2 loss [22] is used for model training.

2.4. Streaming speaker embedding extract

We compare three streaming strategies, denoted M1, M2, and M3. M1 (embedding averaging) simply averages the embedding of the current chunk with those of all prior chunks. M2 (feature concatenation) concatenates all accumulated features before pooling, then passes them through the subsequent network to produce the embedding. M3 (causal retraining) requires retraining the model under a causal architecture, as described below.

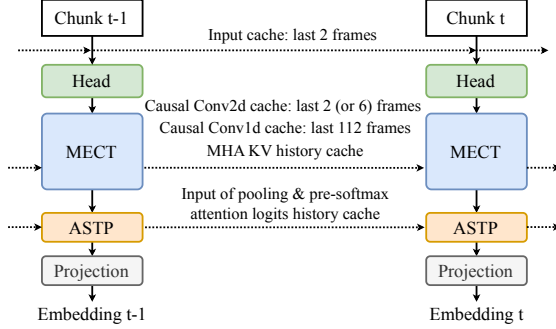


Fig. 2: Streaming inference pipeline of MECT. Illustrating the causal processing and multi-level caching across consecutive chunks.

We first convert the network to a causal architecture. For convolution layers, padding is applied only on the left side of the temporal dimension, and for multi-head attention (MHA), a causal mask restricts each frame to attend only to preceding frames. The model is retrained accordingly.

During streaming inference (Figure 2), no mean normalization is applied to the input fbank features. Conv2d layers cache 2 frames (or 6 for temporal stride 2) for receptive field alignment, while Conv1d layers cache 112 frames to cover the accumulated stride, both concatenated with the current input along the temporal dimension. For MHA, the KV states of all previous steps are concatenated with the current step before attention. For ASTP, the input frame features and intermediate attention logits (feature before softmax) from prior chunks are cached and concatenated with the current chunk to compute global mean and standard deviation. Since only the KV cache and ASTP rely on full history and both are lightweight, the additional cost remains modest. In practice, both caches can be capped to bound memory usage. Our experiments do not apply this cap to focus on validating the streaming algorithm.

3. EXPERIMENT

3.1. Datasets

We evaluated MECT on multiple datasets. For VoxCeleb model training, we used the VoxCeleb2 [23] development set, either alone or combined with VoxBlink2 [24]. SV performance was evaluated with VoxCeleb1-Cleaned protocols: Vox1-O, Vox1-E, Vox1-H and Vox21-val. Additionally, the development sets of CN-Celeb1 and CN-Celeb2 [25] were used for training, with evaluation on CN-Celeb Test. These experiments span different languages, scales, and data sources, validating the effectiveness of the proposed method.

3.2. Training

Experiments were conducted under three data settings. All models were trained in two stages: full training followed by large-margin fine-tuning (LMF) [28]. All experiments used the same hardware and PyTorch environment, and random seeds were fixed across ablation studies to isolate the effect of each component.

VoxCeleb2. VoxCeleb2 was augmented with two-fold speed perturbation (0.9 and 1.1) [29], yielding 17,982 speakers. Two-second segments were randomly extracted with noise and reverberation augmentation via MUSAN [30] and RIR [31] following [26]. SGD was used with momentum 0.9, weight decay $2e-5$, and batch size 512. The learning rate warmed up to 0.6 over 4 epochs, then

Table 2: Evaluation results on different MoE types.

Type	No.E	Vox1-O	Vox1-E	Vox1-H	Vox21-val	Avg	
						EER	DCF
No MoE	—	0.33	0.48	0.85	1.68	0.84	0.064
F-Dense	2	0.34	0.49	0.89	1.71	0.86	0.066
F-Dense	3	0.27	0.49	0.89	1.65	0.83	0.061
F-Dense	4	0.27	0.46	0.85	1.60	0.80	0.059
F-Dense	5	0.35	0.51	0.88	1.61	0.84	0.061
U-Dense	3	0.30	0.47	0.87	1.62	0.82	0.062
U-Dense	4	0.36	0.48	0.89	1.60	0.83	0.067
F-Top1	4	0.36	0.48	0.88	1.61	0.83	0.064
U-Top1	4	0.37	0.48	0.87	1.61	0.83	0.061
F-Top4	6	0.33	0.49	0.89	1.57	0.82	0.066
F-Top4	8	0.30	0.48	0.87	1.55	0.80	0.063
F-Top4	16	0.31	0.47	0.86	1.57	0.80	0.061
F-Top4	32	0.33	0.48	0.88	1.51	0.80	0.063

decayed to $4e-4$ at epoch 120. The SphereFace2 margin was set to 0.0 for 20 epochs, raised to 0.2 over the next 20, and held constant. In the LMF stage, speed perturbation was removed (5,994 speakers), segment length increased to 6 s, the margin was fixed at 0.3, and the learning rate was reduced to $2e-4$ for 7 epochs.

CN-Celeb. The same training recipe was adopted with a maximum learning rate of 1.6. The model from epoch 50 of the first stage was used for LMF, with a learning rate of $8e-4$.

VoxCeleb2 & VoxBlink2. Speed perturbation was not applied. The first-stage training followed the same settings as the VoxCeleb2 experiment, while the second stage used VoxCeleb2 alone for LMF.

3.3. Evaluation

The performance was evaluated using Equal Error Rate (EER) and minimum Decision Cost Function (minDCF) with $C_{FA} = C_{Miss} = 1$, $P_{target} = 0.01$, except on Vox21-val $P_{target} = 0.05$. AS-Norm [32] and QMF [28] were applied for selected systems using VoxCeleb2 development set as backend.

3.4. Streaming inference test

During streaming evaluation, the input audio was processed chunk by chunk, and the final embedding was obtained after the last chunk. For M1 and M2, adjacent chunks overlap by half the chunk size for better performance.

4. RESULTS

Table 2 reports representative results from our full ablation study of different MoE types and expert counts, all trained on VoxCeleb2 and evaluated on the VoxCeleb1 test sets. “No MoE” denotes the baseline without MoE modules. Among all configurations, Frame-level Dense MoE with 4 experts achieves the best results, outperforming the baseline across all test sets with relative gains of 4.7% and 7.8% on average EER and minDCF. Notably, this improvement comes at a negligible parameter cost: the baseline contains 9.40M parameters versus 9.57M for the best MoE configuration.

Table 3 compares MECT with publicly available models on the VoxCeleb1 protocols, where MECT uses frame-level Dense MoE with 4 experts. In group 2, MECT-A2 outperforms ReDimNet2-B4 across all three test sets with the fewest parameters and MACs, achieving particularly notable gains on Vox1-H (9.4% and 5.9% relative improvements on EER and minDCF). In group 3, MECT-B2 surpasses ReDimNet2-B6 with fewer parameters and comparable

Table 3: Evaluation results on VoxCeleb1-Cleaned. * indicates results obtained using the VoxCeleb2 and VoxBlink2 datasets for training. GMACs were measured on 2-s segments counting only the backbone, with ReDimNet2 using the official open-source model.

Model	Params	GMACs	LMF	AS-Norm	Vox1-O EER(%)	Vox1-O minDCF	Vox1-E EER(%)	Vox1-E minDCF	Vox1-H EER(%)	Vox1-H minDCF
ECAPA (C=512) [1, 8]	6.2M	1.04	×	✓	0.94	0.092	1.21	0.129	2.20	0.205
CAM++ [10, 26]	7.2M	1.15	✓	×	0.71	0.109	0.85	0.095	1.66	0.165
ReDimNet2-B3 [16]	2.58M	2.58	✓	×	0.42	0.038	0.66	0.068	1.22	0.120
MECT-A1	3.78M	3.02	✓	×	0.44	0.038	0.60	0.060	1.05	0.108
Gemini DF-ResNet114 [9]	6.5M	5.0	×	✓	0.69	0.067	0.86	0.097	1.49	0.144
ReDimNet2-B4 [16]	4.52M	4.48	✓	×	0.37	0.040	0.58	0.060	1.07	0.102
MECT-A2	4.12M	4.11	✓	×	0.37	0.032	0.54	0.056	0.97	0.096
ResNet293 [26]	28.6M	28.10	✓	×	0.53	0.057	0.71	0.072	1.30	0.127
ECAPA2 [11]	27.1M	187.0	✓	×	0.44	0.041	0.62	0.066	1.15	0.114
WavLM Large [27]	319.3M	28.5	✓	✓	0.37	0.059	0.50	0.055	1.01	0.099
MECT-B1	8.26M	10.35	✓	×	0.37	0.029	0.50	0.051	0.94	0.092
ReDimNet2-B6 [16]	10.45M	13.11	✓	×	0.29	0.027	0.52	0.050	0.99	0.104
MECT-B2	9.57M	14.43	✓	×	0.27	0.024	0.46	0.048	0.85	0.082
W2V-BERT 2.0* [17]	587M	57.90	✓	×	0.14	0.020	0.31	0.032	0.73	0.071
+AS-Norm & QMF			✓	✓	0.12	0.025	0.27	0.028	0.55	0.051
MECT-B2*	9.57M	14.43	✓	×	0.23	0.013	0.29	0.028	0.54	0.052
+AS-Norm & QMF			✓	✓	0.22	0.012	0.28	0.026	0.52	0.048

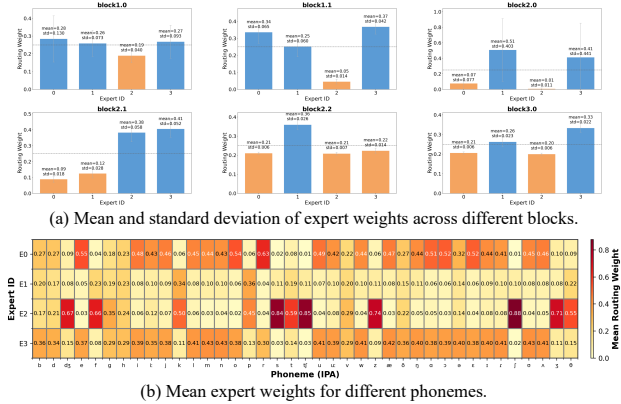


Fig. 3: Statistics of expert weights across network blocks and phonemes.

MACs, delivering consistent improvements on all sets and a 14.1% relative gain on Vox1-H minDCF, while the smaller MECT-B1 already edges past ReDimNet2-B6. With VoxBlink2 training, MECT-B2 outperforms w2v-bert2.0 using far fewer resources, achieving substantial improvements on Vox1-O, Vox1-E, and Vox1-H with relative gains of 10.3% and 24.4% on average EER and minDCF. After AS-Norm and QMF, MECT-B2 further improves and reaches state-of-the-art performance, with minDCF of 0.012, 0.026, and 0.048. Table 4 reports results on CN-Celeb with models trained only on CN-Celeb. MECT-B2 uses frame-level sparse Top-4 MoE with 8 experts and achieves relative improvements of 6.7% on EER and 7.5% on minDCF over ReDimNet2-B6 trained under the same settings.

Figure 3 shows the MoE routing behavior in MECT-B2 on VoxCeleb1. Using the Allosaurus [33] phoneme recognizer, we correlate routing weights with phoneme categories in the first block. As shown in Figure 3(b), E2 assigns high weights (0.65–0.88) to fricatives and affricates [34] (e.g., /s, z, ʃ, f, θ, tʃ, dʒ/), whose high-frequency noise distinguishes them from other phonemes, while vowels (e.g., /e, o, u, æ/) are routed mainly to E0 with moderate weights (0.40–0.55). This specialization emerges without phoneme supervision, suggesting the MoE mechanism naturally captures the acoustic structure of speech.

Table 4: Evaluation results on CN-Celeb.

Model	Params	GMACs	LMF	AS-Norm	CN-Celeb Test EER(%)	DCF
CAM++ [35]	7.2M	1.15	✓	×	6.30	0.370
ERes2NetV2 [35]	17.8M	12.60	✓	×	6.04	0.362
ResNet221 [26]	23.9M	21.29	✓	✓	5.66	0.330
ReDimNet2-B6	12.3M	13.05	✓	×	5.22	0.321
MECT-B2	9.81M	14.46	✓	×	4.87	0.297
w/o MoE	9.57M	14.43	✓	×	5.01	0.303

Table 5: Streaming inference results on VoxCeleb1. MECT-B2[†] denotes the causally retrained model.

Model	Method	Chunk	Vox1-O	Vox1-E	Vox1-H
MECT-B2	Offline	—	0.27	0.46	0.85
	Stream M1	0.1s	12.34	12.36	17.40
		0.2s	4.50	4.45	6.87
		1.0s	0.47	0.67	1.15
	Stream M2	0.1s	7.50	7.55	10.91
		0.2s	2.04	2.14	3.50
		1.0s	0.39	0.58	1.03
MECT-B2 [†]	Offline	—	0.38	0.59	1.05
	Stream M3	0.1s	0.38	0.60	1.06

As shown in Table 5, at a chunk size of 1.0s, M2 achieves the best streaming performance. However, at 0.1s, M1 and M2 degrade severely to 12.34% and 7.50% on Vox1-O, while M3 remains at 0.38%, 0.60%, and 1.06% on Vox1-O, Vox1-E, and Vox1-H, closely matching its offline baseline. This demonstrates the effectiveness of causal retraining for short-chunk streaming.

5. CONCLUSIONS

We proposed MECT, a MoE-based speaker verification model. Four MoE structures were investigated, and performance gains over the baseline were observed across datasets. MECT achieves state-of-the-art performance on VoxCeleb1. A causal retraining strategy enables streaming inference at 0.1s chunks with near-offline accuracy, substantially outperforming non-causal streaming methods. Visualization of expert routing indicates that MoE learns phoneme-level specialization to some extent, without explicit supervision.

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